The Medium

A Pre-Geometric Substrate Theory — Foundation Definition

Bradley K. DePuy, Independent Researcher

ORCID: 0009-0001-0328-2299

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1 Definition

The Medium is a pre-geometric substrate with no beginning and no end. Its sole primitive is ι: an act of self-distinction. Each ι event produces a binary character ($+1$ or -1) — the difference the distinction makes. Character is not held by ι before the event; it is produced by the event.

ι has multiple relational modes (qualitatively distinct aspects of the event, not binary opposition) and intrinsic cohesive attraction. From these two properties, self-organization follows without instruction.

The substrate field records distinction characters across the relational web:

$$\phi : M \to \{+1, -1\} \tag{1}$$

Two ι events of identical character cannot be directly co-manifested (adjacency constraint). This is a logical necessity, not an imposed rule.

2 Scale Inputs

The framework has exactly two inputs:

Input	**Value**	**Role**
Iota scale	$l_\iota = 1.616 \times 10^{-35}$ m	Substrate grain
Hubble scale	$R_H = c/H_0$ (current epoch)	Longest negotiation mode

In Planck units: $G = 1$, $c = 1$, $\hbar = 1/(2\pi)$. What SI physics calls $\hbar$ is the measurement of one projected ι event in the observer's unit system — a conversion artifact, not a primitive constant.

The negotiation energy quantum from the geometric mean of the two scale energies:

$$\varepsilon = \left(E_{\text{Planck}}^2 \cdot E_{\text{Hubble}}\right)^{1/3} = 312.8 \text{ MeV} = m_{\text{proton}}/3 \tag{2}$$

(with Barbero-Immirzi correction $\gamma \approx 0.2375$)

3 Dynamics

The negotiation field N_{ij} encodes the boundary state between adjacent ι events i and j. Lagrangian in Planck units:

$$\mathcal{L} = \tfrac{1}{2}\left(\frac{\partial N}{\partial t}\right)^2 - \tfrac{1}{2}\, N^\top L\, N \tag{3}$$

Hamiltonian (via Legendre transform, $\Pi = \partial N/\partial t$):

$$\hat{H} = \tfrac{1}{2}\Pi^2 + \tfrac{1}{2}\, N^\top L\, N \tag{4}$$

Energy eigenvalues: $E_k = \varepsilon\,\lambda_k$, where λ_k are eigenvalues of the graph Laplacian L.

Stable particles satisfy the closure condition $L \cdot N = 0$. The minimum closed graph is the three-ι triangle ($\lambda = 3$):

$$m_{\text{proton}} = 3\varepsilon = 938.3 \text{ MeV} \tag{5}$$

4 Derived Results

From two inputs and the single correction factor $\gamma \approx 0.2375$:

Quantity	Derived	Measured	Agreement
m_{proton}	$3\varepsilon = 938.3$ MeV	938.3 MeV	Exact
G	$H_0^2/4\pi\rho_\Lambda$	6.674×10^{-11} SI	94–97%
$\hbar$	$m_p c\,k(\gamma)\,r_p/6\pi$	1.055×10^{-34} J·s	99.1%
H_0	68.6 km/s/Mpc	67.4–73.0 (tension range)	Within tension
Λ	$\approx 10^{-122}$ Planck	$\approx 10^{-122}$ Planck	Same order

The ratio $\rho_{\text{Planck}}/\rho_\Lambda \approx 10^{123}$ is the depth of the Medium — the dynamic range from iota grain to cosmic horizon — not a fine-tuning problem.

Analytic prediction: $\Omega_\Lambda = 2/3 \approx 0.667$ (observed: 0.685; discrepancy 2.7\

All three residual gaps (in G, m_{proton}, and $\hbar$) are closed by a single Barbero-Immirzi calculation on the negotiation graph. This is the primary open calculation in the framework.

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6 Document Series and Cross-References

This is Document 1 of 5 in The Medium Framework series.

Doc	File	Content
1	medium_doc1_foundation.org	*This paper*
2	medium_doc2_unified.org	Negotiation Modes, Particle Structure, Proton Mass
3	medium_doc3_lagrangian.org	Lagrangian of the Negotiation Field; QFT connection; SU(3)
4	medium_doc4_hamiltonian.org	Hamiltonian, energy spectrum, canonical quantization, ADM
5	medium_doc5_G_H0.org	Deriving G and H_0 from substrate primitives